THIVAKAR SP

SOFTWARE ENGINEER • IT UNDERGRADUATE

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PROFILE

Results-driven Software Engineer with a strong foundation in Object-Oriented Programming, Data Structures, and System Architecture. Experienced in designing scalable, high-throughput distributed systems using Java, Go, Spring Boot, and AWS. Demonstrated exceptional analytical skills as a Knight on LeetCode with 600+ DSA problems solved. Passionate about writing unit-tested code, adhering to agile SDLC, and leveraging modern practices to drive development efficiency.

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, Go, JavaScript, SQL, HTML/CSS
- **Core CS:** Data Structures & Algorithms, System Architecture, OOP, DBMS, Operating Systems
- **Frameworks & Cloud:** Spring Boot, React.js, REST APIs, Kafka, Redis, AWS, Docker, Git
- **Engineering Practices:** Component Design, System Assembly, Unit Testing, Technical Documentation, Agile SDLC

PROJECTS

Async Notification Fanout Engine

Go, PostgreSQL, Redis, Kafka

- Executed component design for a distributed notification system using Go and Kafka to reliably process 10,000+ Email, SMS, and Push alerts daily.
- Integrated Redis to process events with sub-millisecond latency and ensure strict idempotency, preventing duplicate notifications.
- Applied testing frameworks and added Prometheus metrics to monitor the system's performance, ensuring 99.9% uptime under peak traffic.

Distributed API Rate Limiter

Java, Spring Boot, Redis, AWS

- Implemented a scalable rate-limiting service using Java and Spring Boot to manage and control up to 5,000 incoming API requests per second.
- Utilized token bucket and sliding window algorithms, applying robust concurrency controls to safely restrict user request limits across distributed nodes.
- Leveraged Redis for fast data processing (¡5ms latency), containerized with Docker, and deployed the application to AWS.

CodeTrail (LeetCode to GitHub Sync)

JavaScript, Chrome APIs, REST API

- Built a browser extension that automatically synchronizes solved LeetCode problems directly to a GitHub repository, reducing manual sync time by 100%.
- Extracted and saved critical problem metrics (execution time, language, difficulty), while maintaining detailed, automated technical documentation.
- Managed background tasks via Manifest V3 service workers to handle network retries and API limits smoothly without impacting browser rendering performance.

ACHIEVEMENTS & CERTIFICATIONS

- **Competitive Programming:** Knight @LeetCode (leetcode.com/u/Thivakar_Parthiban); Solved 600+ Data Structures and Algorithms problems.
- **Hackathons:** Participated in Smart India Hackathon (SIH), Amazon HackOn, Adobe India Hackathon, MSME Hackathon, and SAP Hackathon.
- **Certifications:** Microsoft Certified: SQL AI Developer Associate (DP-800), AWS Certified Cloud Practitioner, Nutanix Multicloud Infrastructure (MCI), Nutanix Certified Professional Network and Security (NCP-NS), HackerRank SQL Advanced.

EDUCATION

B.TECH IN INFORMATION TECHNOLOGY — 2023 – 2027
Sri Krishna College of Engineering and Technology — *CGPA: 7.5*